

Course Name: Machine Learning

Course Code: AM 623, AM623D

Duration : 01 Hour

Max Marks : 10

1. Suppose you ran logistic regression twice, one with $\lambda=0$, and $\lambda=1$, one of the times you got parameters $\theta = [91.37, 25.34]^T$, and other times you got $\theta = [20.17, 2.16]^T$. However, you forgot, which value of θ corresponds to which value of λ . which one θ do you think corresponds to $\lambda=0$ [2.5 M]
2. Suppose you train a logistic regression classifier $h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$. Suppose $\theta_0 = -6$, $\theta_1 = 0$, $\theta_2 = 1$. Sketch the decision boundary found by the classifier. [2.5 M]
3. Learn a decision tree for the dataset shown below with the depth of the tree=1 [5M]

Humidity	Wind	PlayTennis
High	Weak	No
High	Strong	No
High	Weak	Yes
High	Weak	Yes
Normal	Weak	Yes
Normal	Strong	No
Normal	Strong	Yes
High	Weak	No
Normal	Weak	Yes
Normal	Weak	Yes
Normal	Strong	Yes
High	Strong	Yes

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END SEMESTER - Examination (Spring, 2022)

Assessment: 3. Course Name: Machine Learning, Course Code: AM 623, AM623D

1. Write the hypothesis of an SVM classifier? What are support vectors in SVM, and what is its significance? Whether SVM a nonlinear classifier? [2 M]
2. How multi-label classification is performed using a binary classifier? Suppose if you are asked to perform 5 class classification, how many SVM classifiers are needed to perform the task? [2 M]
3. Explain how Regression is performed using Neural Networks (NN)? Explain how stochastic gradient descent differs from gradient descent? [2 M]
4. Classify the sample [Outlook= Rain, Temperature= Hot, Humidity= High, Wind= Strong] based on data using the Naïve Bayes [4M]

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

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END SEMESTER - Examination (Spring, 2022)

Course Name: Machine Learning
Duration : 03 Hours

Course Code: AM 623/623D
Max Marks : 50

Note:

1. Marks are indicated against each question. Write clearly and legibly. All diagrams are to be neatly labelled. Calculators are allowed.
2. Attempt **ALL** questions.
3. Before attempting please ensure you have the correct question paper.

Q1	a)	When Machine Learning is feasible?	2M																				
	b)	Explain the terms: overfitting, underfitting, and the right fit of the model with diagrams, and explain the criteria for detecting them. Explain any two techniques that address overfitting.	3M																				
	c)	Explain Feature Normalization? How does it impact the accuracy of the model in the case of linear regression and K-Nearest Neighbor classifier?	3M																				
	d)	Explain the Pearson correlation coefficient (r). Compute the r value for Data A and Data B. Based on the r-value, which algorithm (Linear regression, Polynomial Regression) do you prefer to apply for the following data sets? <u>Data A:</u> <table border="1"> <tr> <td>x (input)</td> <td>1</td> <td>2</td> <td>4</td> <td>5</td> </tr> <tr> <td>y (output)</td> <td>2.5</td> <td>4</td> <td>5</td> <td>7</td> </tr> </table> <u>Data B:</u> <table border="1"> <tr> <td>x (input)</td> <td>-2</td> <td>-1</td> <td>1</td> <td>2</td> </tr> <tr> <td>y (output)</td> <td>4</td> <td>1</td> <td>1</td> <td>4</td> </tr> </table>	x (input)	1	2	4	5	y (output)	2.5	4	5	7	x (input)	-2	-1	1	2	y (output)	4	1	1	4	2M
x (input)	1	2	4	5																			
y (output)	2.5	4	5	7																			
x (input)	-2	-1	1	2																			
y (output)	4	1	1	4																			
Q2	a)	Explain Linear regression based on gradient descent and Normal equations.	5M																				
	b)	Use locally weighted linear regression for predicting the value $x=3.5$ for the Data A ? (Refer Q 1d.). Consider two nearest neighbors, bandwidth parameter= $1/\sqrt{2}$	3M																				
	c)	Explain the Coefficient of Determination (R^2), and compute the R^2 value for Hypothesis: $h(x)=1.62+x$, considering Data A (Refer Q 1d.).	2M																				

Q3	a)	Suppose you train a logistic regression classifier $h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$, where g : sigmoid function, $\theta_0 = -6$, $\theta_1 = 2$, $\theta_2 = 3$. Sketch the decision boundary found by the classifier. By applying classifier $h_{\theta}(x)$ on the data points: A(1,1), B(-1,1), C(2,1), D(4,-1), mention which data points belong to the same category.	4M
	b)	Suppose if you learnt a SVM classifier ($W^T x + b = 0$) with $W = [-3, 2, 1, -1]$, and $b = 2$ for classifying the data $A = [1, 1, -1, 1]$, $B = [-1, 2, 2, -1]$, and $C = [2, -1, -1, 2]$, $D = [3, 5, 7, 2]$. Mention which data points belong to the same category.	3M
	c)	Define the performance metrics- Accuracy, Sensitivity, Specificity, F1-Score, for the below data set.	3M

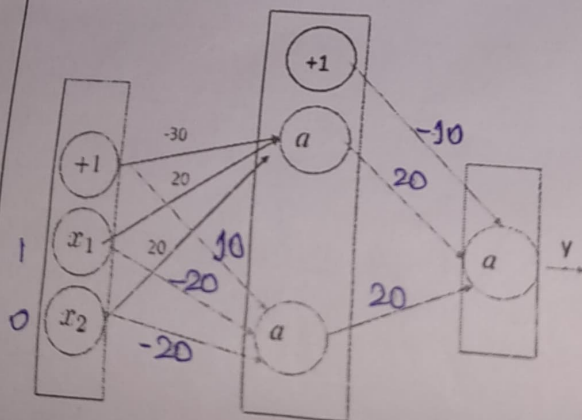
y	1	1	0	1	0	0	1	1	1	0	0	1	0	1
$\hat{y}$	1	0	0	0	0	1	1	0	1	0	0	1	1	1

y: Ground truth, $\hat{y}$: Predicted

Q4 a) Learn the Decision Tree for the following data:

Instance	A1	A2	Classification
1	T	T	+
2	T	T	+
3	T	F	-
4	F	F	+
5	F	T	-
6	F	T	-

b) In the following neural network, what will be the output 'y' obtained for different combinations of inputs x_1 and x_2 (Here x_1 and x_2 take binary inputs-0,1. 'a' represents the sigmoidal activation function., Numbers in the figure represents the learned weights)



	c)	Explain any two techniques that address the problem of gradient descent stuck at local minima with respect to Neural Networks.	2M												
Q5	a)	Explain Bayes Theorem with all terminologies	2M												
	b)	Apply K- means clustering algorithm with K=2 on the data [20, 2, 5, 7, 10, 25, 12, 18]. Use 20 and 25 as starting cluster centers.	3M												
	c)	Use the KNN algorithm for finding the value at x=3.5 (Use K=2) <table><tr><td>x (input)</td><td>1</td><td>2</td><td>4</td><td>5</td><td>8</td></tr><tr><td>y (output)</td><td>2.5</td><td>4</td><td>5</td><td>7</td><td>6.5</td></tr></table>	x (input)	1	2	4	5	8	y (output)	2.5	4	5	7	6.5	2M
x (input)	1	2	4	5	8										
y (output)	2.5	4	5	7	6.5										
	d)	Explain how the AUC-ROC curve is plotted and its significance in model selection.	3M												

***** ALL THE BEST *****

Subject: Machine Learning: Spring 2023

Assessment No. 1

Marks: 10

1. Considering a Hypothesis: $H_{\theta}(x) = \theta_0 + \theta_1 x$, with $\alpha=0.01$, $\theta_0 = \theta_1 = 0$, for the dataset. Suppose the values of (x^i, y^i) are: (1, 1), (2, 3), (3, 5), (4, 7), (5, 9). Using these values, what are the updated values of θ_0 and θ_1 after three iterations of gradient descent?
4M
2. Explain how stochastic gradient descent (SGD) differs from gradient descent. Give two advantages of SGD over gradient descent
3M
3. Explain the term Coefficient of Determination, and how the model performance is accessed based on it.
3M

$$\theta_1 = 0.88$$
$$\theta_0 = 0.1723$$

Assessment 3

Sub: Machine Learning: (AM623/AM623D)

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1. For an SVM with a decision boundary of the form:  $W^T x + b = 0$ . The support vectors are  $x_1 =$ ,  $x_2 =$ ,  $x_3 =$ , ,  $y_1 = y_2 = -1$ ,  $y_3 = 1$ . Compute  $W$  and  $b$ .

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$\alpha_1 = 0.2; \alpha_2 = 0.5 \quad \alpha_3 = 1 \quad y_3 = 1 \quad (3 \text{ M})$$
$$y_1 = y_2 = -1$$

2. Explain (a) : Under what circumstances do we choose Soft margin SVM?  
(b) : How SVM can be used for Linearly non-separable data? (2 M)

3. How to realize a Logistic regression classifier based on Neural Network (NN) (2.5 M)

4. Explain the difference in the NN architecture for Regression and Classification (2.5 M)

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$$6.2x_1 + 7.9x_2 = -0.7$$

$$x_2 = -0.112$$
$$-0.08$$



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Note:

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2. Attempt **ALL** questions.
3. Before attempting please ensure you have the correct question paper.

|            |     |                                                                                                                                                                                                                                                                                                                                                                                |           |      |   |   |   |            |     |     |     |      |  |
|------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------|---|---|---|------------|-----|-----|-----|------|--|
| Q1         | a)  | Explain in detail what is needed for applying the machine learning algorithms to any given problem                                                                                                                                                                                                                                                                             | 2M        |      |   |   |   |            |     |     |     |      |  |
|            | b)  | With respect to Regression- explain the terms overfitting, underfitting, and the right fit of the model with diagrams. Explain any two techniques that can address the problem of underfitting.                                                                                                                                                                                | 5M        |      |   |   |   |            |     |     |     |      |  |
|            | c)  | What is feature normalization and why it is needed? How the Data: [10, 7, 15, 74, 99, 25] transforms after applying mean normalization                                                                                                                                                                                                                                         | 3M        |      |   |   |   |            |     |     |     |      |  |
| Q2         | a)  | Explain Polynomial regression in detail, and on what basis, Polynomial regression will be chosen over linear regression                                                                                                                                                                                                                                                        | 5M        |      |   |   |   |            |     |     |     |      |  |
|            | b)  | Explain the Coefficient of Determination ( $R^2$ ), and compute the $R^2$ value for Hypothesis: $h(x)=2x+3$ , considering the below data.                                                                                                                                                                                                                                      | 5M        |      |   |   |   |            |     |     |     |      |  |
|            |     | <table border="1"> <tr> <td>x (input)</td> <td>1</td> <td>2</td> <td>3</td> <td>5</td> </tr> <tr> <td>y (output)</td> <td>4.8</td> <td>7.2</td> <td>8.6</td> <td>12.8</td> </tr> </table>                                                                                                                                                                                      | x (input) | 1    | 2 | 3 | 5 | y (output) | 4.8 | 7.2 | 8.6 | 12.8 |  |
| x (input)  | 1   | 2                                                                                                                                                                                                                                                                                                                                                                              | 3         | 5    |   |   |   |            |     |     |     |      |  |
| y (output) | 4.8 | 7.2                                                                                                                                                                                                                                                                                                                                                                            | 8.6       | 12.8 |   |   |   |            |     |     |     |      |  |
| Q3         | a)  | Suppose you train a logistic regression classifier $h_\theta(x)=g(\theta_0+ \theta_1x_1+ \theta_2x_2+ \theta_3x_3)$ , where $g$ : sigmoid function, $\theta_0=-6$ , $\theta_1=2$ , $\theta_2=3$ , $\theta_3=5$ . By applying classifier $h_\theta(x)$ on the data points: A (1,1, 1), B(-1,1,-1), C(2,1, 1), D(4,-1,0), mention which data points belong to the same category. | 5 M       |      |   |   |   |            |     |     |     |      |  |
|            | b)  | Explain in detail the working of the Hard margin SVM Classifier                                                                                                                                                                                                                                                                                                                | 5M        |      |   |   |   |            |     |     |     |      |  |
| Q4         | a)  | Draw the Decision Tree for the following Boolean functions                                                                                                                                                                                                                                                                                                                     | 5 M       |      |   |   |   |            |     |     |     |      |  |
|            | i)  | $A \wedge \neg B$                                                                                                                                                                                                                                                                                                                                                              |           |      |   |   |   |            |     |     |     |      |  |
|            | ii) | $A \vee [B \wedge C]$                                                                                                                                                                                                                                                                                                                                                          |           |      |   |   |   |            |     |     |     |      |  |



|    |    |                                                                                                                                                                                                                                                                             |     |
|----|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|    | b  | Give out any two reasons why classification based on Neural Network is preferred over logistic regression.                                                                                                                                                                  | 5 M |
| Q5 | a) | Explain in detail how Naïve Bayes classifier works?                                                                                                                                                                                                                         | 5 M |
|    | b) | Apply K- means clustering algorithm with $K=2$ on the data [10, 25, 3, 10, 17, 12, 40, 64, 14, 52]. Use 25 and 40 as starting cluster centers. Mention separately the data points of two clusters and the cluster centroid. To which cluster the data <u>30</u> belongs to? | 5 M |

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ALL THE BEST

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## Assessment No 1

Sub: Machine Learning  
Answer All Questions

Feb 13<sup>th</sup> 2024

1. Refer to below Table.  $x, y$  indicates the independent and dependent variables respectively. Fit a regression model assuming a  $h_{\theta}(x) = \theta_0 + \theta_1 \cdot x$ . Assume a learning parameter  $\alpha = 0.1$ ,  $\theta_0 = 4.5$  and  $\theta_1 = -1.5$  for gradient descent algorithm. Repeat the gradient descent algorithm for three iterations. Report the (i)  $\theta$  parameters (ii) Mean square error, after three iterations. **(5 Marks)**

| $x$ | $y$ |
|-----|-----|
| -2  | 9.5 |
| -1  | 6.8 |
| 1.5 | 3   |
| 3   | 1   |

2. Compute the Coefficient of Determination for the model obtained in Q. 1 ( obtained after running the Gradient descent algorithm for three iterations ) **(3 Marks)**
3. True challenge for Machine Learning algorithms “*Is to solve the tasks that are easy for people to perform but hard for people to describe formally*”. How do you justify the above statement? Give two concrete reasons **(2 Marks)**

Q.1) Classify the following data  $A = [2, 3]$ ,  $B = [-1, 2]$ ,  $C = [1, -2]$ ,  $D = [1, 2]$  based on the category it belongs. The decision boundary separating the classes is given by  $2x_1 - x_2 - 3 = 0$ . What is the probability that the sample  $D$  belongs to the +ve category (3M)

Q.2) If the data  $[-1, 2]$  and  $[5, 10]$  are the closest points belonging to different classes. Derive the expression for the decision boundary using KNN classifier (Euclidean distance as a similarity measure)

Q.3)  $TP=7$ ,  $FN=3$ ,  $FP=4$ ,  $TN=6$ , Compute F1 score.

Q.4) Explain how AUC-ROC is plotted & what it signifies.



**Subject: Machine Learning AM623**  
**Assessment No 3**

1. Build a decision tree for the following data

4M

| Instance | <del>A1</del> | A2 | A3 | classification |
|----------|---------------|----|----|----------------|
| 1        | F             | F  | F  | -              |
| 2        | F             | F  | T  | -              |
| 3        | F             | T  | F  | +              |
| 4        | F             | T  | T  | -              |
| 5        | T             | F  | F  | +              |
| 6        | T             | F  | T  | +              |
| 7        | T             | T  | F  | +              |
| 8        | T             | T  | T  | -              |

2. If  $x_1 = [1, -2, 4]$ ,  $x_2 = [-1, 3, 2]$ ,  $x_3 = [2, 3, -1]$ ,  $\alpha_1 = 0.2$ ,  $\alpha_2 = 2$ ,  $\alpha_3 = 7$ ,  $y_1 = y_2 = 1$ ,  $y_3 = -1$ . Compute the value of  $w$  (in the context of SVM Classifier)
3. Suppose you receive an email and your spam filter classifies it as spam. You know that the spam filter is 99% accurate in classifying spam emails correctly. Out of all the emails you receive 2% are actually spam. What is the probability that an email classified as spam by the filter is actually spam?

3M

3M

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2. Attempt **ALL** questions.
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| Q1                | a)                | <p>If <math>h_1</math> and <math>h_2</math> are the two hypothesis functions learnt using an ML algorithm for the same dataset with the following errors (same data split is used for training and testing)</p> <table><thead><tr><th><math>h_1</math></th><th><math>h_2</math></th></tr></thead><tbody><tr><td>Train Error: 10 %</td><td>Train Error: 13 %</td></tr><tr><td>Test Error: 18 %</td><td>Test Error: 15 %</td></tr></tbody></table> <p>Explain which Hypothesis function (<math>h_1</math> or <math>h_2</math>) will you choose for your application with appropriate technical terminologies.</p> | $h_1$ | $h_2$ | Train Error: 10 % | Train Error: 13 % | Test Error: 18 % | Test Error: 15 % | 5M |
|-------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------------------|-------------------|------------------|------------------|----|
|                   | $h_1$             | $h_2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |       |                   |                   |                  |                  |    |
| Train Error: 10 % | Train Error: 13 % |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |       |                   |                   |                  |                  |    |
| Test Error: 18 %  | Test Error: 15 %  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |       |                   |                   |                  |                  |    |
|                   | b)                | <p>Give the mathematical formulation of Lasso and Ridge Regression and explain how specifically these formulations will result in addressing the challenges of overfitting.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 | 5 M   |       |                   |                   |                  |                  |    |

| Q2         | a)  | <p>Is it possible to obtain a linear decision boundary using the Logistic Regression? If it so, give me a generic mathematical expression for the linear decision boundary learnt using the logistic regression.</p>                                                                                                                  | 5M        |    |   |   |   |            |     |      |    |    |
|------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----|---|---|---|------------|-----|------|----|----|
|            | b)  | <p>Compute the Mean square error and coefficient of determination for Hypothesis: <math>h(x)=5x+2</math>, considering the below data.</p> <table><thead><tr><th>x (input)</th><th>1</th><th>2</th><th>3</th><th>5</th></tr></thead><tbody><tr><th>y (output)</th><td>6.5</td><td>12.8</td><td>16</td><td>28</td></tr></tbody></table> | x (input) | 1  | 2 | 3 | 5 | y (output) | 6.5 | 12.8 | 16 | 28 |
| x (input)  | 1   | 2                                                                                                                                                                                                                                                                                                                                     | 3         | 5  |   |   |   |            |     |      |    |    |
| y (output) | 6.5 | 12.8                                                                                                                                                                                                                                                                                                                                  | 16        | 28 |   |   |   |            |     |      |    |    |



|                 |    |                                                                                                                                                                                                                                                                                                                                                                                                      |              |   |   |   |   |   |   |   |   |   |   |   |   |                 |   |   |   |   |   |   |   |   |   |   |   |   |
|-----------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|---|---|---|---|---|---|---|---|---|---|---|-----------------|---|---|---|---|---|---|---|---|---|---|---|---|
| Q3              | a) | Draw the Decision Tree for the following Boolean functions using Gini impurity (Hint: Create a dataset by tabulating the truth table by taking different combinations of 0 and 1 for A and B.)<br><br>$A \vee [B \wedge C]$<br><br>( $\wedge$ represents AND, $\vee$ represents OR, $-$ represents COMPLIMENT or NEGATION operator)                                                                  | 5 M          |   |   |   |   |   |   |   |   |   |   |   |   |                 |   |   |   |   |   |   |   |   |   |   |   |   |
|                 | b) | Compute Precision, Recall, F1 score and Accuracy for the below Table. <table><tr><td>Actual Class</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>Predicted Class</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr></table> | Actual Class | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Predicted Class | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |
| Actual Class    | 0  | 0                                                                                                                                                                                                                                                                                                                                                                                                    | 0            | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |   |   |   |                 |   |   |   |   |   |   |   |   |   |   |   |   |
| Predicted Class | 1  | 0                                                                                                                                                                                                                                                                                                                                                                                                    | 0            | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |   |   |   |                 |   |   |   |   |   |   |   |   |   |   |   |   |

|    |    |                                                                                                                                                                                                                                                                                 |     |
|----|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Q4 | a) | Explain the need for kernel methods in the SVM classifier. Name any two kernels along with their mathematical formulation.                                                                                                                                                      | 5 M |
|    | b) | Suppose if you learnt a SVM classifier ( $W^T x + b = 0$ ) with $W = [-3, 2, 1, -1]$ , and $b = 2$ for classifying the data $A = [1, 1, -1, 1]$ , $B = [-1, 2, 2, -1]$ , and $C = [2, -1, -1, 2]$ , $D = [3, 5, 7, 2]$ . Mention which data points belong to the same category. | 5 M |

|    |    |                                                                                                                                                                                                                                                                              |     |
|----|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Q5 | a) | Explain in detail the working of DBSCAN algorithm and how it overcomes the limitations of K Means algorithm.                                                                                                                                                                 | 5 M |
|    | b) | Apply K- means clustering algorithm with $K = 2$ on the data $[10, 25, 3, 10, 17, 12, 30, 40, 64, 14, 52]$ . Use 3 and 30 as starting cluster centers. Mention separately the data points of two clusters and the cluster centroid. To which cluster the data 33 belongs to? | 5 M |

END

$$\begin{bmatrix} 1 & 2 & 5 & 6 & 7 \\ 7 & 3 & 6 & 4 & 2 \\ 9 & 2 & 5 & 8 & 4 \\ 8 & 6 & 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 2 & 4 & 7 \\ 1 & 2 & 1 \\ 3 & 5 & 4 \end{bmatrix}$$

4x5                      5x3

$$\begin{bmatrix} \phantom{0} & \phantom{0} & \phantom{0} \\ \phantom{0} & \phantom{0} & \phantom{0} \\ \phantom{0} & \phantom{0} & \phantom{0} \\ \phantom{0} & \phantom{0} & \phantom{0} \end{bmatrix}$$

4x3

## Practice

Q1

$\lambda$  penalises the cost function, and therefore results in some lowering of the parameters values.

When  $\lambda = 0$ , it means there was no regularization, while  $\lambda = 1$  ensures some regularization. So, if I have to make an informed guess, then I would say,

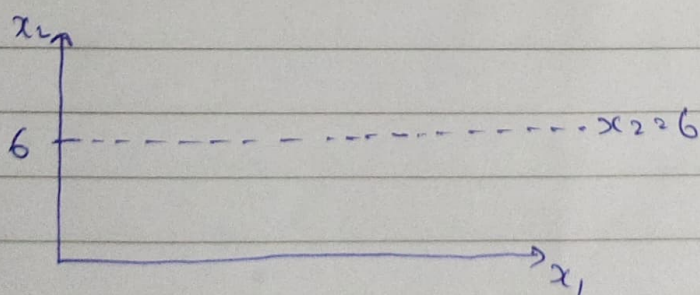
$$\theta = [91.37, 25.34]^T \text{ belongs to } \lambda = 0.$$

Q2

$$\theta_0 = -6, \theta_1 = 0, \theta_2 = 1,$$

$$h_\theta(x) = g(-6 + 0x_1 + x_2)$$

$\therefore$  The decision boundary will be:  $-6 + 0x_1 + x_2 = 0$   
 $\Rightarrow x_2 = 6$





Q3 Let the dataset be named as  $S$ .  $|S| = 12$

+ve = 8, -ve = 4

$$\text{Entropy}(S) = -\frac{4}{12} \log\left(\frac{4}{12}\right) - \frac{8}{12} \log\left(\frac{8}{12}\right)$$

$$= 0.9183$$

Humidity

Let  $S_1$  be the dataset when Humidity = High

$S_1$

| Humidity | Wind   | Play Tennis |
|----------|--------|-------------|
| High     | Weak   | No          |
| High     | Strong | No          |
| High     | Weak   | Yes         |
| High     | Weak   | Yes         |
| High     | Weak   | No          |
| High     | Strong | Yes         |

$$\# |S_1| = 6, \text{ +ve} = 3, \text{ -ve} = 3$$

$$\text{Entropy}(S_1) = -\frac{3}{6} \log\left(\frac{3}{6}\right) - \frac{3}{6} \log\left(\frac{3}{6}\right)$$

$$\text{Entropy}(S_1) = 1$$



Let  $S_2$  be the dataset when Humidity = Normal

$S_2$

| Humidity | Wind   | Play Tennis |
|----------|--------|-------------|
| Normal   | Weak   | Yes         |
| Normal   | Strong | No          |
| Normal   | Strong | Yes         |
| Normal   | Weak   | Yes         |
| Normal   | Weak   | Yes         |
| Normal   | Strong | Yes         |

$$|S_2| = 6, \text{ +ve} = 5, \text{ -ve} = 1$$

$$\text{Entropy}(S_2) = -\frac{5}{6} \log\left(\frac{5}{6}\right) - \frac{1}{6} \log\left(\frac{1}{6}\right)$$

$$\text{Entropy}(S_2) = 0.6500$$

$$\begin{aligned} \text{Gain}(S, \text{Humidity}) &= \text{Entropy}(S) - \sum_{\substack{v \in \text{Values} \\ (\text{Humidity})}} \frac{|S_v|}{|S|} \text{Entropy}(S_v) \\ &= \text{Entropy}(S) - \frac{|S_1|}{|S|} \text{Entropy}(S_1) - \frac{|S_2|}{|S|} \text{Entropy}(S_2) \\ &= 0.9183 - \frac{6}{12} \cdot 1 - \frac{6}{12} \cdot 0.6500 \end{aligned}$$

$$\boxed{\text{Gain}(S, \text{Humidity}) = 0.0933}$$



Wind

let  $S_1$  be the dataset when Wind = Weak

$S_1$

| Humidity | Wind | Play Tennis |
|----------|------|-------------|
| High     | Weak | No          |
| High     | Weak | Yes         |
| High     | Weak | Yes         |
| Normal   | Weak | Yes         |
| High     | Weak | No          |
| Normal   | Weak | Yes         |
| Normal   | Weak | Yes         |

$$|S_1| = 7, \text{ true} = 5, \text{ false} = 2$$

$$\text{Entropy}(S_1) = -\frac{5}{7} \log\left(\frac{5}{7}\right) - \frac{2}{7} \log\left(\frac{2}{7}\right)$$

$$\text{Entropy}(S_1) = 0.8631$$

let  $S_2$  be the dataset when Wind = Strong

$S_2$

| Humidity | Wind   | Play Tennis |
|----------|--------|-------------|
| High     | Strong | No          |
| Normal   | Strong | No          |
| Normal   | Strong | Yes         |
| Normal   | Strong | Yes         |
| High     | Strong | Yes         |



$$|S_2'| = 5, +ve = 3, -ve = 2$$

$$\text{Entropy}(S_2') = -\frac{3}{5} \log\left(\frac{3}{5}\right) - \frac{2}{5} \log\left(\frac{2}{5}\right)$$

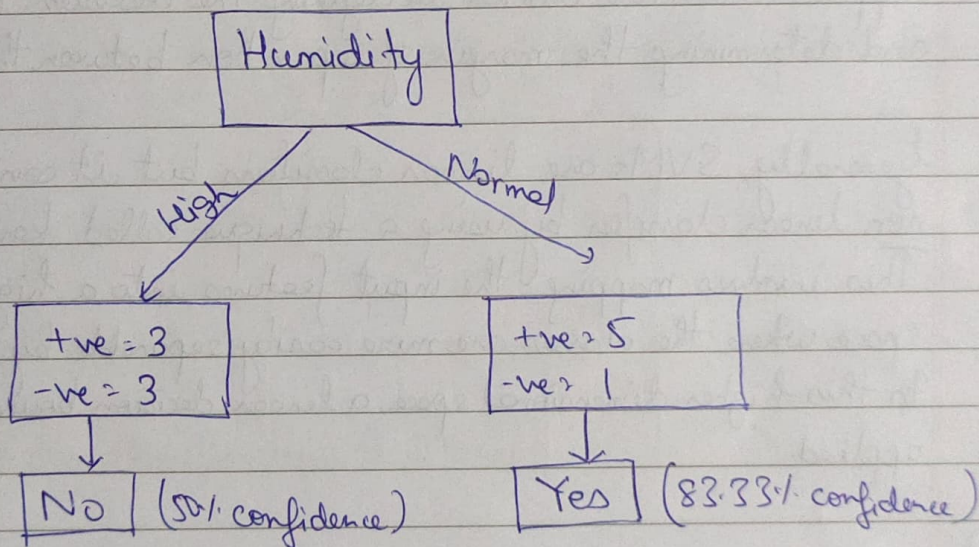
$$= 0.9709$$

$$\text{Gain}(S, \text{Wind}) = \text{Entropy}(S) - \frac{|S_1'|}{|S|} \text{Entropy}(S_1') - \frac{|S_2'|}{|S|} \text{Entropy}(S_2')$$

$$= 0.9183 - \frac{7}{12} \cdot 0.8631 - \frac{5}{12} \cdot 0.9709$$

$$\text{Gain}(S, \text{Wind}) = 0.0102$$

$$\text{Gain}(S, \text{Humidity}) > \text{Gain}(S, \text{Wind})$$





### Assessment - 3

Q1

The hypothesis of a SVM classifier is given by:-

$$h(x) = \text{sign}(W^T x + b)$$

where,

$W$  is weight vector

$b$  is bias term

$x$  is feature vector of input data.

Support vectors in SVM are the data points that lie closest to the decision boundary and influence its position. These support vectors are crucial in defining the decision boundary and determining the margin of separation between the classes.

Generally, SVMs are linear classifiers but it can be a non linear classifier by using a technique called kernel trick. This involves mapping the input features into a higher dimensional space where the classes are more easily separable by a hyperplane. In this higher dimensional space, a linear decision boundary can be applied.

Q2

Let's say I have five labels namely  $L = (A, B, C, D, E)$ .  
I have two ways to perform multi-label classification using binary classifiers:



## (i) One vs One

I train a binary classifier for each of the following labels:  
 $(l_i, l_j) \forall l_i, l_j \in L, i \neq j$

I will have  $\frac{k(k-1)}{2}$  classifiers for  $k$  label classifications.

Then, I will see which classifiers are giving polarised output (output ~~with~~ with high confidence).

I will pick that label as the label of the test instance.

## (ii) One vs Rest

I train a binary classifier for each of the following ~~test~~ labels:  
 $(l_i, \sim l_i) \forall l_i \in L$

I will have  $k$  classifiers for  $k$  label classification.

Then, I will see which classifier is giving higher positive ~~for~~ output.

For a five class classification, I will need:

(i) 20 SVMs for One vs One

(ii) 5 SVMs for One vs Rest.

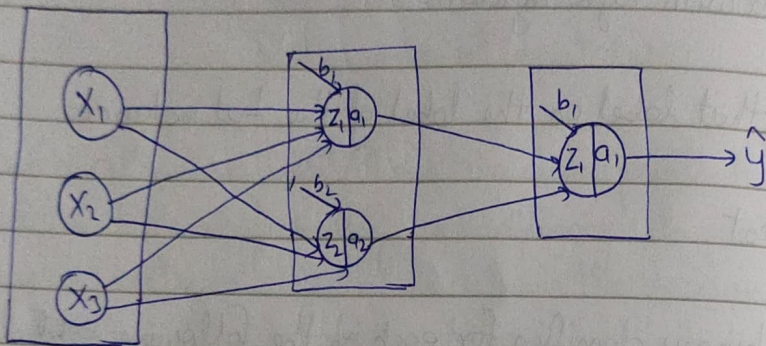


Q3

By having appropriate activation functions, we can perform Regression using Neural Networks.

Let's say I am trying to come up with the ~~estimation~~ a House Price Predictor based on the following features:

$$f(x_1, x_2, x_3) \rightarrow Y$$



$$\text{Then, } W^{[1]} = \begin{bmatrix} \omega_{11} & \omega_{12} & \omega_{13} \\ \omega_{21} & \omega_{22} & \omega_{23} \end{bmatrix}_{2 \times 3}$$

$$A^{[0]} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_{3 \times 1} \quad B^{[0]} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}_{2 \times 1}$$

$$Z^{[1]} = W^{[1]} A^{[0]} + B^{[0]}$$

$$Z^{[1]} = \begin{bmatrix} \omega_{11}x_1 + \omega_{12}x_2 + \omega_{13}x_3 + b_1 \\ \omega_{21}x_1 + \omega_{22}x_2 + \omega_{23}x_3 + b_2 \end{bmatrix}$$

The activations are ReLU for ~~both~~ all neurons,



$$A^{[1]} = \text{Relu}(Z^{[1]})$$

$$A^{[1]} = \begin{bmatrix} \text{ReLU}(w_{11}x_1 + w_{12}x_2 + w_{13}x_3 + b_1) \\ \text{ReLU}(w_{21}x_1 + w_{22}x_2 + w_{23}x_3 + b_2) \end{bmatrix}_{2 \times 1}$$

$$W^{[2]} = \begin{bmatrix} w'_{11} & w'_{12} \end{bmatrix}_{1 \times 2} \quad B^{[1]} = \begin{bmatrix} b'_1 \end{bmatrix}_{1 \times 1}$$

$$Z^{[2]} = W^{[2]} A^{[1]} + B^{[1]}$$

$$A^{[2]} = \text{ReLU}(Z^{[2]}) = \hat{y}$$

To learn all  $w$ 's and  $b$ 's, we can backpropagate using MSE loss and gradient descent algorithm.

Following are the differences between SGD and GD:

| SGD                                                                | GD                                                                 |
|--------------------------------------------------------------------|--------------------------------------------------------------------|
| (i) Takes single training sample at <del>one</del> each iteration. | (i) Takes all training samples at <del>one</del> each iteration.   |
| (ii) Noisier error vs iteration curve.                             | (ii) Smooth error vs iteration curve.                              |
| (iii) Computationally simpler.                                     | (iii) Computationally difficult when no. of samples are very high. |



Q4

$h_1$  :- Play Tennis (Yes)  
 $h_2$  :- Not Play Tennis (No)

$D = [D_1 = \text{Rain}, D_2 = \text{Hot}, D_3 = \text{High}, D_4 = \text{Strong}]$   
 $H = \{h_1, h_2\}$

$h_{\text{MAP}} = ?$

$$h_{\text{MAP}} = \arg \max_{h_k \in H} \{P(h_k|D)\}$$

$$P(h_1|D) = \frac{P(D|h_1)P(h_1)}{P(D)}$$

$$P(h_2|D) = \frac{P(D|h_2)P(h_2)}{P(D)}$$

$$h_{\text{MAP}} = \arg \max_{h \in \{h_1, h_2\}} \left\{ \frac{P(D|h_1)P(h_1)}{P(D)}, \frac{P(D|h_2)P(h_2)}{P(D)} \right\}$$

Assuming  $D_i$ 's are independent.

$$h_{\text{MAP}} = \arg \max_{h \in \{h_1, h_2\}} \left\{ \frac{P(h_1) \cdot P(D_1|h_1)P(D_2|h_1)P(D_3|h_1)P(D_4|h_1)}{P(h_2)P(D_1|h_2)P(D_2|h_2)P(D_3|h_2)P(D_4|h_2)} \right\}$$



| Outlook ( $D_1$ ) | $D_2$ | $D_3$ | $D_4$ | Play Tennis |
|-------------------|-------|-------|-------|-------------|
| - Sunny           | H     | H     | W     | N           |
| - Sunny           | H     | H     | S     | N           |
| + Overcast        | H     | H     | W     | Y           |
| + Rain            | M     | H     | W     | Y           |
| + Rain            | C     | N     | W     | Y           |
| - Rain            | C     | N     | S     | N           |
| + Overcast        | C     | N     | S     | Y           |
| - Sunny           | M     | H     | W     | N           |
| + Sunny           | C     | N     | W     | Y           |
| + Rain            | M     | N     | W     | Y           |
| + Sunny           | M     | N     | S     | Y           |
| + Overcast        | M     | H     | S     | Y           |
| + Overcast        | H     | N     | W     | Y           |
| - Rain            | M     | H     | S     | N           |

$$P(h_1) = P(Y) = 9/14, \quad P(h_2) = P(N) = 5/14$$

$$P(D_1|h_1) = 3/9, \quad P(D_2|h_1) = 2/9, \quad P(D_3|h_1) = 3/9, \quad P(D_4|h_1) = 3/9$$

$$P(h_2) = P(N) = 5/14$$

$$P(D_1|h_2) = 2/5, \quad P(D_2|h_2) = 2/5, \quad P(D_3|h_2) = 4/5, \quad P(D_4|h_2) = 3/5$$

$$h_{MAP} = \arg \max_h \left\{ \frac{9}{14} \times \frac{3}{9} \times \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9}, \frac{5}{14} \times \frac{2}{5} \times \frac{2}{5} \times \frac{4}{5} \times \frac{3}{5} \right\}$$

$$= \arg \max_h \{ 0.005291, 0.027429 \}$$

$$h_{MAP} = h_2$$

So, we can classify the sample as Not play tennis with a confidence of

$$\frac{0.027429}{0.005291 + 0.027429} \times 100\% = 83.83\%$$



## End Sem (Spring 2022)

Q1

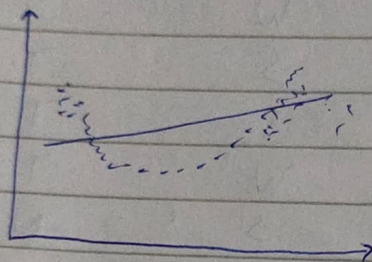
(a) Machine learning is feasible when:

(i) we have enough data

(ii) data shows some pattern which can be expressed mathematically

(b)

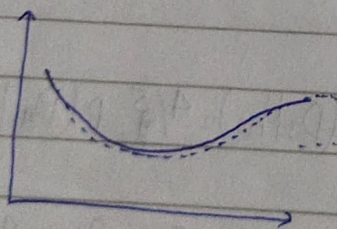
Underfitting



Training Error  $\uparrow$  (high bias)

Test error  $\uparrow$  (high variance)

Right fit

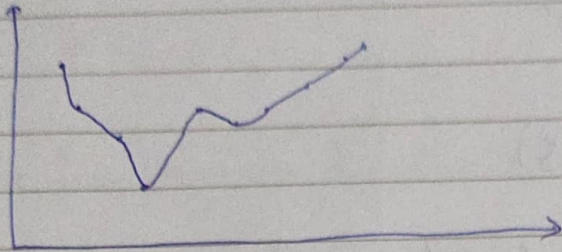


Training Error  $\downarrow$  (low bias)

Test error  $\downarrow$  (low variance)



## Overfitting



Training Error ↓ (low bias)  
Testing Error ↑ (high variance).

Following are the two techniques to address overfitting:

- (i) Manually dropping features.
- (ii) Regularization.

(c) Feature normalization is the process of scaling features such that their values end up ranging between 0 and 1. It is also known as Min-Max scaling.

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

It removes the impact any feature might have had because of its value from different scale. Normalizing features for distance based algorithms improve their accuracy.



(d)

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}}$$

Data A ( $\bar{X}=3, \bar{Y}=4.625$ )

| $X_i$ | $Y_i$ | $(X - \bar{X})$ | $(Y - \bar{Y})$ | $(X - \bar{X})^2$ | $(Y - \bar{Y})^2$ | $(X - \bar{X})(Y - \bar{Y})$ |
|-------|-------|-----------------|-----------------|-------------------|-------------------|------------------------------|
| 1     | 2.5   | -2              | -2.125          | 4                 | 4.515625          | 4.250                        |
| 2     | 4     | -1              | -0.625          | 1                 | 0.390625          | 0.625                        |
| 4     | 5     | 1               | 0.375           | 1                 | 0.140625          | 0.375                        |
| 5     | 7     | 2               | 2.375           | 4                 | 5.640625          | 4.750                        |
|       |       |                 |                 | 10                | 10.687500         | 10.000                       |

$$r = \frac{10}{\sqrt{10 \times 10.6875}} = 0.9673$$

Data B ( $\bar{X}=0, \bar{Y}=2.5$ )

| $X_i$ | $Y_i$ | $(X - \bar{X})$ | $(Y - \bar{Y})$ | $(X - \bar{X})(Y - \bar{Y})$ | $(X - \bar{X})^2$ | $(Y - \bar{Y})^2$ |
|-------|-------|-----------------|-----------------|------------------------------|-------------------|-------------------|
| -2    | 4     | -2              | 1.5             | -3                           | 4                 | 2.25              |
| -1    | 1     | -1              | -1.5            | +1.5                         | 1                 | 2.25              |
| 1     | 1     | 1               | -1.5            | -1.5                         | 1                 | 2.25              |
| 2     | 4     | 2               | 1.5             | +3                           | 4                 | 2.25              |
|       |       |                 |                 | 0                            | 10                | 9                 |

$$r = \frac{0}{\sqrt{10 \times 9}} = 0$$

Linear Regression applies to Data A & Polynomial Regression applies to Data B



Q2

a) Let's say I have following dataset:

$$R(X, y)$$

The linear regression hypothesis is:

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

### Gradient Descent

We want to learn  $\theta_0$  and  $\theta_1$  such that,

$$L(\theta) = \frac{1}{2m} \sum_{i=1}^m (y^{(i)} - h_{\theta}(x^{(i)}))^2 \text{ is minimum.}$$

$$L(\theta) = \frac{1}{2m} \sum_{i=1}^m (y^i - \theta_0 - \theta_1 x^i)^2$$

$$\frac{\partial L(\theta)}{\partial (\theta_0)} = \frac{1}{m} \sum_{i=1}^m (y^i - \theta_0 - \theta_1 x^i) (-1)$$

$$\frac{\partial L(\theta)}{\partial (\theta_1)} = \frac{1}{m} \sum_{i=1}^m (y^i - \theta_0 - \theta_1 x^i) (-x^i)$$

The parameters are updated using the following update rule:

$$\theta_0 \leftarrow \theta_0 - \alpha \frac{\partial L(\theta)}{\partial (\theta_0)}$$

$$\theta_1 \leftarrow \theta_1 - \alpha \frac{\partial L(\theta)}{\partial (\theta_1)}$$



Normal equations

$$\text{Let } X = \begin{bmatrix} 1 & x^{(1)} \\ 1 & x^{(2)} \\ 1 & x^{(3)} \\ \vdots & \vdots \\ 1 & x^{(m)} \end{bmatrix}_{m \times (\dim(x)+1)} \quad Y = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(m)} \end{bmatrix}_{m \times 1}$$

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}_{2 \times 1}$$

We know that  $\hat{Y} = X\theta$

$$E = (Y - X\theta)$$

we want to minimize  $\sum E^2$  which is given by  $E^T E$

$$E^T E = (Y - X\theta)^T (Y - X\theta)$$

$$= (Y^T - \theta^T X^T) (Y - X\theta)$$

$$E^T E = (Y^T Y - Y^T X\theta - \theta^T X^T Y + \theta^T X^T X\theta)$$

$$E^T E = (Y^T Y - 2Y^T X\theta + \theta^T X^T X\theta)$$

Since, we want to minimize  $E^T E$  wrt  $\theta$ .

$$\frac{\partial E^T E}{\partial \theta} = 0$$

$$\frac{\partial Y^T Y}{\partial \theta} = 0$$

$$\frac{\partial (-2Y^T X\theta)}{\partial \theta} = (-2Y^T X)^T = -2X^T Y$$

$$\frac{\partial (\theta^T X^T X\theta)}{\partial \theta} = 2X^T X\theta$$



$$-2X^T Y + 2X^T X \theta = 0$$

$$2X^T X \theta = 2X^T Y$$

$$\theta = (X^T X)^{-1} X^T Y$$

$$\therefore \hat{\theta}_{OLS} = (X^T X)^{-1} X^T Y$$

(c)

Coefficient of Determination is a metric which tells us how much of the variance in data is accounted for by our hypothesis or model.

$$R^2 = 1 - \frac{\sum (Y - \hat{Y})^2}{\sum (Y - \bar{Y})^2}$$

$$\hat{Y} = 1.62 + x$$

| x | y   | $\hat{y}$ | $(y - \hat{y})$ | $(y - \bar{y})$ | $(y - \bar{y})^2$ | $(y - \hat{y})^2$ |
|---|-----|-----------|-----------------|-----------------|-------------------|-------------------|
| 1 | 2.5 | 2.62      | -0.12           | -2.8125         | 4.515625          | 0.0144            |
| 2 | 4   | 3.62      | 0.38            | -0.625          | 0.390625          | 0.1444            |
| 4 | 5   | 5.62      | -0.62           | 0.375           | 0.140625          | 0.3844            |
| 5 | 7   | 6.62      | 0.38            | 2.375           | 5.640625          | 0.1444            |
|   |     |           |                 |                 | 10.6875           | 0.6876            |

$$R^2 = 1 - \frac{0.6876}{10.6875} = 0.9356$$

$R^2 \approx 1$ , good model (ideal case)

$R^2 \approx 0$ , baseline model

$R^2 < 0$ , bad model.

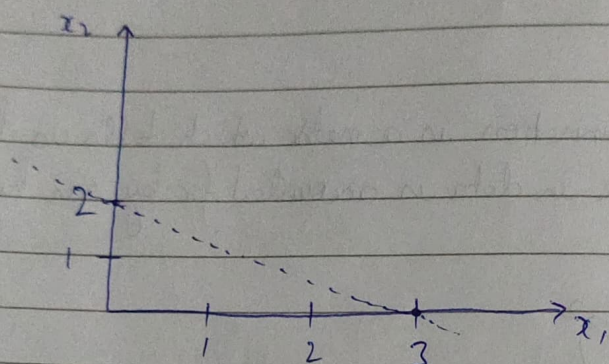


Q3

(a)  $\theta_0 = -6, \theta_1 = 2, \theta_2 = 3$

$$h_\theta(x) = g(-6 + 2x_1 + 3x_2)$$

The decision boundary is the line :-  
 $2x_1 + 3x_2 = 6$



$$h_\theta(A) = \text{sign}(g(-6 + 2(1) + 3(1)))$$

$$h_\theta(A) = -$$

$$h_\theta(B) = \text{sign}(-6 - 2 + 3)$$

$$h_\theta(B) = -$$

$$h_\theta(C) = \text{sign}(-6 + 4 + 3)$$

$$h_\theta(C) = +$$

$$h_\theta(D) = \text{sign}(-6 + 8 - 7)$$

$$h_\theta(D) = -$$

A, B, D belong to same category.



$$(b) h(x) = \text{sign}(W^T x + b)$$

$$h(A) = \text{sign}\left([-3, 2, 1, -1] \begin{bmatrix} 1 \\ 1 \\ -1 \\ 1 \end{bmatrix} + 2\right)$$

$$= \text{sign}(-3 + 2 - 1 - 1 + 2)$$

$$h(A) = -$$

$$h(B) = \text{sign}([-3, 2, 1, -1] [-1, 2, 2, -1] + 2)$$

$$= \text{sign}(12)$$

$$h(B) = +$$

$$h(C) = \text{sign}([-3, 2, 1, -1] [2, -1, -1, 2] + 2)$$

$$= \text{sign}(-6 - 2 - 1 - 2 + 2)$$

$$h(C) = -$$

$$h(D) = \text{sign}([-3, 2, 1, -1] [3, 5, 7, 2] + 2)$$

$$= \text{sign}(-9 + 10 + 7 - 2 + 2)$$

$$h(D) = +$$

$\{A, C\}, \{B, D\}$

(c)

|                    |            |   |   |    |
|--------------------|------------|---|---|----|
|                    | $\hat{y}$  | 0 | 1 |    |
| $y$                | $\nwarrow$ |   |   |    |
| TN $\rightarrow$ 0 |            | 4 | 3 | FN |
| FP $\rightarrow$ 1 |            | 2 | 5 |    |
|                    |            |   |   | TP |

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{9}{14}$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{5}{7}$$

$$\text{Sensitivity / TPR / Recall} = \frac{TP}{TP + FN} = \frac{5}{8}$$

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{4}{6}$$



$$F1 \text{ score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Q2  
(b)

We know that

$$w^{(i)} = e^{-\left(\frac{(x^{(i)} - x_q)^2}{2\tau^2}\right)}$$

$$x_q = 3.5 \quad \tau = \frac{1}{\sqrt{2}}, \quad 2\tau^2 = 1$$

$$w^{(1)} = e^{-((1-3.5)^2)} = e^{-6.25}$$

$$w^{(2)} = e^{-((2-3.5)^2)} = e^{-2.25}$$

$$w^{(3)} = e^{-((4-3.5)^2)} = e^{-0.25}$$

$$w^{(4)} = e^{-((5-3.5)^2)} = e^{-2.25}$$

$$Y_{x_q} = \frac{e^{-0.25} \cdot 5 + e^{-2.25} \cdot 7}{e^{-0.25} + e^{-2.25}}$$

$$Y_{x_q} = 5.24$$



Q4

(a) Let the dataset be named  $S$ . $S$ 

| Instance | A1 | A2 | Classification |
|----------|----|----|----------------|
| 1        | T  | T  | +              |
| 2        | T  | T  | +              |
| 3        | T  | F  | -              |
| 4        | F  | F  | +              |
| 5        | F  | T  | -              |
| 6        | F  | T  | -              |

~~Ent~~  $|S| = 6$ , +ve: 3, -ve: 3

$$\text{Entropy}(S) = -\frac{3}{6} \log\left(\frac{3}{6}\right) - \frac{3}{6} \log\left(\frac{3}{6}\right)$$

$$\text{Entropy}(S) = 1$$

 $A1$ Let  $S_1$  be the dataset when  $A1 = T$  $S_1$ 

| Instance | A1 | A2 | Classification |
|----------|----|----|----------------|
| 1        | T  | T  | +              |
| 2        | T  | T  | +              |
| 3        | T  | F  | -              |

 $|S_1| = 3$ , +ve: 2, -ve: 1

$$\text{Entropy}(S_1) = -\frac{2}{3} \log\left(\frac{2}{3}\right) - \frac{1}{3} \log\left(\frac{1}{3}\right)$$

$$\text{Entropy}(S_1) = 0.9183$$



let  $S_2$  be the dataset when  $A_1 = F$

$S_2$

| Instance | $A_1$ | $A_2$ | Classification |
|----------|-------|-------|----------------|
| 4        | F     | F     | +              |
| 5        | F     | T     | -              |
| 6        | F     | T     | -              |

$$|S_2| = 3, +ve = 1, -ve = 2$$

$$\text{Entropy}(S_2) = -\frac{2}{3} \log\left(\frac{2}{3}\right) - \frac{1}{3} \log\left(\frac{1}{3}\right)$$

$$= 0.9183$$

$$\text{Gain}(S, A_1) = \text{Entropy}(S) - \frac{|S_1|}{|S|} \text{Entropy}(S_1) - \frac{|S_2|}{|S|} \text{Entropy}(S_2)$$

$$= 1 - \frac{3}{6} \cdot 0.9183 - \frac{3}{6} \cdot 0.9183$$

$$\boxed{\text{Gain}(S, A_1) = 0.0817}$$

$A_2$

Let  $S'_1$  be the dataset when  $A_2 = T$

$S'_1$

| Instance | $A_1$ | $A_2$ | Classification |
|----------|-------|-------|----------------|
| 1        | T     | T     | +              |
| 2        | T     | T     | +              |
| 5        | F     | T     | -              |
| 6        | F     | T     | -              |

$$|S'_1| = 4, +ve = 2, -ve = 2, \text{Entropy}(S'_1) = 1$$



Let  $S_2'$  be the dataset when  $A_2 = F$

$S_2'$

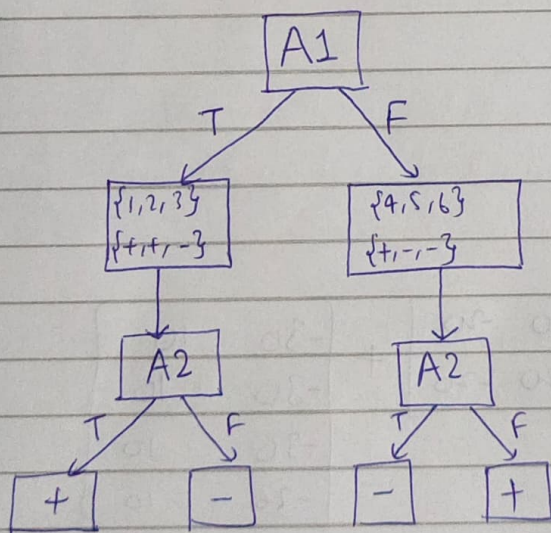
| Instance | $A_1$ | $A_2$ | Classification |
|----------|-------|-------|----------------|
| 3        | T     | F     | -              |
| 4        | F     | F     | +              |

$$|S_2'| = 2, +ve = 1, -ve = 1, \text{Entropy}(S_2') = 1$$

$$\begin{aligned} \text{Gain}(S, A_2) &= \text{Entropy}(S) - \frac{|S_1'|}{|S|} \text{Entropy}(S_1') - \frac{|S_2'|}{|S|} \text{Entropy}(S_2') \\ &= 1 - \frac{4}{6} \cdot 1 - \frac{2}{6} \cdot 1 \end{aligned}$$

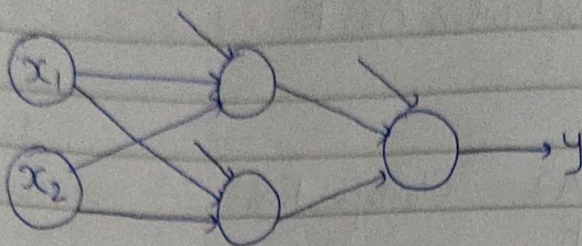
$$\text{Gain}(S, A_2) = 0$$

Since,  $\text{Gain}(S, A_1) > \text{Gain}(S, A_2)$ , we split on  $A_1$





(b)



$$W^{[1]} = \begin{bmatrix} 20 & 20 \\ -20 & -20 \end{bmatrix} \quad B^{[1]} = \begin{bmatrix} -30 \\ 10 \end{bmatrix} \rightarrow \begin{bmatrix} -30 & 10 \\ -30 & 10 \\ -30 & 10 \\ -30 & 10 \end{bmatrix}$$

$$W^{[2]} = \begin{bmatrix} 20 & 20 \end{bmatrix} \quad B^{[2]} = \begin{bmatrix} -10 \end{bmatrix} \rightarrow \begin{bmatrix} -10 \\ -10 \\ -10 \\ -10 \end{bmatrix}$$

~~Q3~~

|  | $x_1$ | $x_2$ |
|--|-------|-------|
|  | 0     | 0     |
|  | 0     | 1     |
|  | 1     | 0     |
|  | 1     | 1     |

$$A^{[0]} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$Z^{[1]} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 20 & -20 \\ +20 & -20 \end{bmatrix} + \begin{bmatrix} -30 & 10 \\ -30 & 10 \\ -30 & 10 \\ -30 & 10 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ +20 & -20 \\ 20 & -20 \\ 40 & -40 \end{bmatrix} + \begin{bmatrix} -30 & 10 \\ -30 & 10 \\ -30 & 10 \\ -30 & 10 \end{bmatrix}$$



$$Z^{[1]} = \begin{bmatrix} -30 & 10 \\ -10 & -10 \\ -10 & -10 \\ +10 & -30 \end{bmatrix}$$

$$A^{[1]} = \sigma(Z^{[1]})$$

$$A^{[1]} = \begin{bmatrix} \sigma(-30) & \sigma(10) \\ \sigma(-10) & \sigma(-10) \\ \sigma(-10) & \sigma(10) \\ \sigma(-30) & \sigma(10) \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$A^{[1]} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$$

$$Z^{[2]} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$$

$$Z^{[2]} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 20 \\ 20 \end{bmatrix} + \begin{bmatrix} -10 \\ -10 \\ -10 \\ -10 \end{bmatrix}$$

$$Z^{[2]} = \begin{bmatrix} 20 \\ 0 \\ 0 \\ 20 \end{bmatrix} + \begin{bmatrix} -10 \\ -10 \\ -10 \\ -10 \end{bmatrix} = \begin{bmatrix} 10 \\ -10 \\ -10 \\ 10 \end{bmatrix}$$



$$y = A^{[2]} \cdot \sigma(Z^{[2]})$$

$$= \begin{bmatrix} \sigma(10) \\ \sigma(-10) \\ \sigma(-10) \\ \sigma(10) \end{bmatrix}$$

$$y = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\therefore$$

| $x_1$ | $x_2$ | $y$ |
|-------|-------|-----|
| 0     | 0     | 1   |
| 0     | 1     | 0   |
| 1     | 0     | 0   |
| 1     | 1     | 1   |

(c) Following are the two techniques that address the problem of gradient descent stuck at local minima with respect to Neural Networks:

- (i) **SGD with Momentum**: It leverages accumulated momentum to navigate through regions of low curvature and small local minima.
- (ii) **Simulated Annealing**: It introduces controlled randomness to explore the solution space and escape from suboptimal solutions.



Q5

(a) According to Bayes Theorem:-

If there are two events A and B, then

$$P(A|B) P(B) = P(B|A) P(A)$$

or

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

where,

 $P(A|B)$  : posterior probability $P(B|A)$  : likelihood $P(A)$  : prior $P(B)$  : marginal (scaling factor)(b) Iteration 1 ( $C_1^0 = 20, C_2^0 = 25$ )  $K=2$ 

| $X_i$ | $d(X_i, C_1^0)$ | $d(X_i, C_2^0)$ | $C_i$ (new) |
|-------|-----------------|-----------------|-------------|
| 20    | 0               | 5               | $C_1$       |
| 2     | 18              | 23              | $C_1$       |
| 5     | 15              | 20              | $C_1$       |
| 7     | 13              | 18              | $C_1$       |
| 10    | 10              | 15              | $C_1$       |
| 25    | 5               | 0               | $C_2$       |
| 12    | 8               | 13              | $C_1$       |
| 18    | 2               | 7               | $C_1$       |

$$C_1^1 = \frac{20+2+5+7+10+12+18}{7}$$

$$= 10.57$$

$$C_2^1 = 25$$



Iteration 2 ( $C_1' = 10.57$ ,  $C_2' = 25$ )

| $X_i$ | $d(X_i, C_1')$ | $d(X_i, C_2')$ | $C$   |
|-------|----------------|----------------|-------|
| 20    | 9.43           | 5              | $C_2$ |
| 2     | 8.57           | 23             | $C_1$ |
| 5     | 5.57           | 20             | $C_1$ |
| 7     | 3.57           | 18             | $C_1$ |
| 10    | 0.57           | 15             | $C_1$ |
| 25    | 14.43          | 0              | $C_2$ |
| 12    | 1.43           | 13             | $C_1$ |
| 18    | 7.43           | 7              | $C_2$ |

$$C_1^2 = \frac{2+5+7+10+12}{5}$$

$$C_2^2 = \frac{20+25+18}{3}$$

$$C_1^2 = 7.2$$

$$C_2^2 = 21$$

Iteration 3 ( $C_1^2 = 7.2$ ,  $C_2^2 = 21$ )

| $X_i$ | $d(X_i, C_1^2)$ | $d(X_i, C_2^2)$ | $C$   |
|-------|-----------------|-----------------|-------|
| 20    | 12.8            | 1               | $C_2$ |
| 2     | 5.2             | 19              | $C_1$ |
| 5     | 2.2             | 16              | $C_1$ |
| 7     | 0.2             | 14              | $C_1$ |
| 10    | 2.8             | 11              | $C_1$ |
| 25    | 17.8            | 4               | $C_2$ |
| 12    | 4.8             | 9               | $C_1$ |
| 18    | 10.8            | 3               | $C_2$ |

Since, there is no change,

$$C_1 = \{2, 5, 7, 10, 12\} \quad C_2 = \{20, 25, 18\}$$

$$C_1 = 7.2, \quad C_2 = 21$$

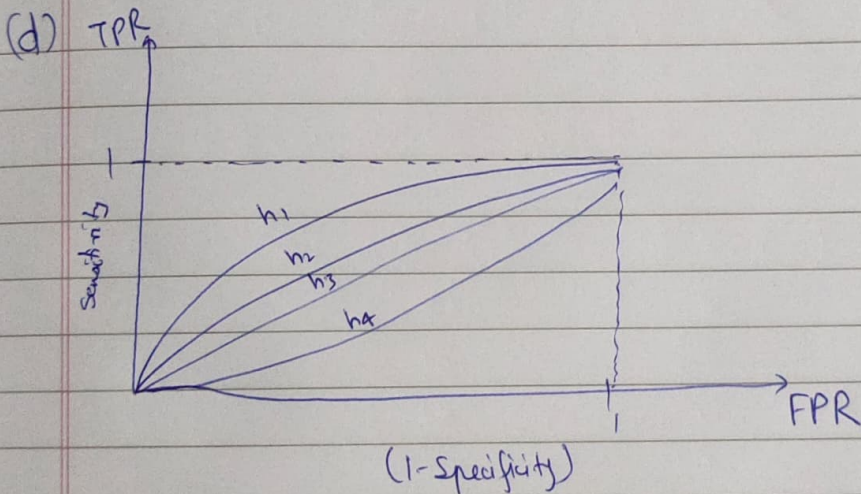


(c)

$$x = 3.5 \quad K = 2$$

| $x^i$ | $d(x, x^i)$ | $y$ |
|-------|-------------|-----|
| 1     | 2.5         | 2.5 |
| 2     | 1.5         | 4   |
| 4     | 0.5         | 5   |
| 5     | 1.5         | 7   |
| 8     | 4.5         | 6.5 |

$$y_x = \frac{5+7}{2} = 6.$$



If  $AUC(h_1) > AUC(h_2)$ , then select  $h_1$

$h_3$  is baseline, so any model  $h_i$  such that  $AUC(h_i) < AUC(h_3)$  is a bad hypothesis.



Assessment-3

Q1

$$x_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad y_1 = -1$$

$$x_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad y_2 = -1$$

$$x_3 = \begin{bmatrix} 5 \\ 6 \end{bmatrix} \quad y_3 = 1$$

Since,  $x_1, x_2, x_3$  are support vectors and  $y_1, y_2, y_3$  are labels

$$w^T x_1 + b = -1$$

$$w^T x_2 + b = -1$$

$$w^T x_3 + b = 1$$

$$\text{let } w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}, \alpha_1 = 0.2, \alpha_2 = 0.5, \alpha_3 = 1$$

$$\begin{aligned} w_1 + 2w_2 + b &= -1 \\ 2w_1 + 3w_2 + b &= -1 \\ 5w_1 + 6w_2 + b &= 1 \end{aligned}$$

$$w = \sum \alpha_i y_i x_i$$

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \alpha_1 y_1 x_1 + \alpha_2 y_2 x_2 + \alpha_3 y_3 x_3$$

$$= \begin{bmatrix} -0.2 \\ -0.4 \end{bmatrix} + \begin{bmatrix} -1 \\ -1.5 \end{bmatrix} + \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 3.8 \\ 4.1 \end{bmatrix}$$



Now,

$$\begin{bmatrix} 3.8 & 4.1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b = -1$$

$$3.8 + 8.2 + b = -1$$

$$12.0 + b = -1$$

$$b = -13$$

$$\therefore w = \begin{bmatrix} 3.8 \\ 4.1 \end{bmatrix}, \quad b = -13$$

Q2

(a) If we want to increase margin while allowing for some misclassifications, then we choose Soft Margin SVM.

(b) Same as Assessment 3 Q1.

Q

Q3 We can have a Neural Network with sigmoid activation at output layer and we can have binary cross entropy loss function to realize logistic regression classifier using Neural Networks.

Q4 We have ReLU activation in Regression while ~~tan~~ tanh or sigmoid activation in Classification. Also, loss function is MSE loss in Regression while Binary CE loss in classifiers.



## Assessment - 2

Q1

$$f(x, y, z) = 3x + 2y - 7z + 10$$

$$f(A) = 3(1) + 2(1) - 7(3) + 10 \\ = -6$$

$$f(B) = 3(1) + 2(1) - 7(1) + 10 \\ = 8$$

$$f(C) = 3(5) + 2(0) - 7(1) + 10 \\ = 18$$

$$f(D) = 3(0) + 2(0) - 7(1) + 10 \\ = 3$$

B, C, D belong to same category.

Q2

A scenario, where there is data imbalance.  
Use Sensitivity, Specificity, F1 score.

Q3

Outliers can lead to overfitting, as model while trying to reduce impurity may further split which was not necessary. Pruning can be performed to address such scenarios.

Q4

- (a) False
- (b) False
- (c) Decision Trees
- (d) False
- (e) False (Not sure)



End Sem (Spring 2023)

Q1

- (a) Same as End Sem (Spring 2022) Q1 (a)
- (b) Same as End Sem (Spring 2022) Q1 (b)

Following are the two techniques to address underfitting:

- (i) Feature engineering
- (ii) Model complexity

- (c) Same as End Sem (Spring 2022) Q1 (c)

Apply  $X' = \frac{X - \mu}{\sigma}$

Q2

- (a) If Pearson correlation coefficient closer to 0, then go for polynomial regression, if  $|r| \approx 1$ , then go for linear regression
- (b) Same as End Sem (Spring 2022)

Q3

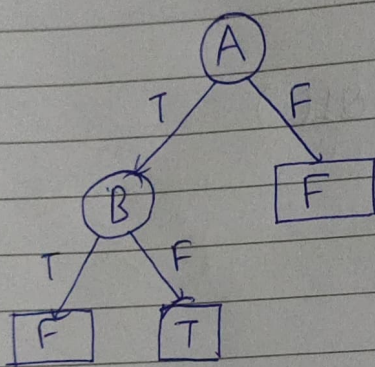
- (a) Same as above
- (b) In Hard Margin SVM classifier, we ensure no misclassification against lower margin.



Q4

(a)(i)  $A \wedge \sim B$

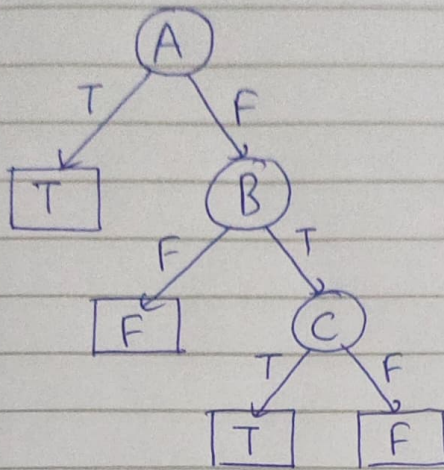
| A | B | $A \wedge \sim B$ |
|---|---|-------------------|
| T | T | F                 |
| T | F | T                 |
| F | T | F                 |
| F | F | F                 |



(ii)  $A \vee (B \wedge C)$

| A | B | C | $A \vee (B \wedge C)$ |
|---|---|---|-----------------------|
| T | T | T | T                     |
| T | T | F | T                     |
| T | F | T | T                     |
| T | F | F | T                     |
| F | T | T | T                     |
| F | T | F | F                     |
| F | F | T | F                     |
| F | F | F | F                     |





E

(b) Following are two reasons why Neural Network based classifiers are preferred over logistic regression:-

- (i) Complexity of Relationships: Neural Networks can capture complex, non-linear relationships between features and output classes more effectively than logistic regression.
- (ii) Feature learning: Neural Networks have the capability to automatically learn relevant features from the raw data, whereas logistic regression relies on manually engineered features.

Q5

(a) Given dataset, and a set of hypotheses, it tells us which hypothesis is more likely to occur based on query data sample using Bayes Theorem.

$$h_{MAP} = \arg \max_{h \in H} \{ P(h | X_q) \}$$

(b) Same as above Ed Sem (Spring 2022)